

Degree: B.Tech., Stream: ALL (Group- A)

Term – I, First Semester Examination, August - 2025

Subject Code: ESCCS102

Subject Name: Introduction to AI and Data Science using Python

Full Marks: 30

Duration: 1 Hour

Part - A

Attempt only five questions

Each question carries 2 marks

1. Write the syntax of a for loop in Python. 2
2. Briefly discuss about when to use while loop and when to use for loop. 2
3. Write the differences between list and tuple. 2
4. What is the output of the following code and why? 2

```
>> print(type("10") == type(10))
```
5. Solve the following code and write the output. Justify your answer. 2

```
>> while ((2*2==4) and not(2+2==5) and (2*2==2+2)) :  
    print(" Hello World! ")
```
6. Solve the following code and write the output. Justify your answer. 2

```
>> temp = 30  
>> print("Low" if temp<25 else "Medium" if temp<35 else "High")
```

Part - B

Attempt only two questions

Each question carries 5 marks

7. Write a python program to print all the prime numbers in a range taken from the user. 5
8. Discuss the local and global variable with suitable example. 5
9. For a user input integer n, print the following pattern 5

```

      *
    *  *
  *   *  *
*    *  *  *
* *   *  *  *
* *  *  *  *  *

```

..... up to n levels

Part - C

Attempt only one question

Each question carries 10 marks

10. Write a Python program that performs the following: 10
- Prompts the user to enter the name of a geometric shape: Triangle, Square, Rectangle, or Circle.
 - Based on the input, the program should:
 - Ask the user for the required dimensions (e.g., sides, radius, etc.)
 - Calculate both the perimeter and the area of the chosen shape.
 - Display the results with appropriate labels.
11. Create a basic calculator in Python that performs the following operations like Addition, Subtraction, Multiplication and Division. 10

The calculator should:

- Accept multiple numbers as input from the user (not limited to two).
- Allow the user to choose the operation to perform.
- Implement each operation using a separate function.
- Handle division by zero gracefully with a proper error message.
- Display the result after the chosen operation.

Note: Use only basic constructs such as for loops, if-else conditions, lists, and functions. Avoid using any built-in functions like sum() or eval().